

## Quick Activity 2: Canteen Group Order

A group of classmates eats at the canteen and wants to split the bill equally. Create an algorithm that computes the total bill and the amount each person pays.

### Decomposition

Assume the costs are positive numbers and the number of classmates is a whole number.

| INPUT                                   | PROCESS                                                                           | OUTPUT                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------|
| Food cost, drink cost, number of people | Add food and drink cost to get total bill. Divide total bill by number of people. | Total bill and amount each person pays |

### Pseudocode

START

```
READ food, drinks, people
totalBill = food + drinks
payPerPerson = totalBill / people
DISPLAY totalBill
DISPLAY payPerPerson
```

END

## Dry Run Calculations

### Test Case 1

Input:

- Food = 250
- Drinks = 50
- People = 3

| STEP | VARIABLE       | CALCULATIONS | SOLUTIONS |
|------|----------------|--------------|-----------|
| 1    | Total Bill     | $250 + 50$   | ₱300      |
| 2    | Pay Per Person | $300 / 3$    | ₱100      |

### Output

Total Bill = ₱300

Pay Per Person = ₱100

### Test Case 2

Input:

- Food = 420
- Drinks = 180
- People = 4

| STEP | VARIABLE       | CALCULATIONS | SOLUTIONS |
|------|----------------|--------------|-----------|
| 1    | Total Bill     | $420 + 180$  | ₱600      |
| 2    | Pay Per Person | $600 / 4$    | ₱150      |

### Output

Total Bill = ₱600

Pay Per Person = ₱150